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SCENES



Games and programs will often need to present different scenes to the user. For the purpose of this discussion, I'll define a *scene* as any window layout and related user interactions that are significantly different from any other. For example, a game like *Space Invaders* may have a starting or *splash* scene, a main game play scene, a high scores scene, and perhaps an ending or goodbye scene.

In this chapter I will discuss two different approaches to writing a program that has multiple scenes. First, I'll introduce the state machine technique, which works well for relatively small programs. Then I'll show a fully object-oriented approach where each scene is implemented as an object, under the control of a scene manager. The latter is much more scalable for larger programs.

The State Machine Approach

At the beginning of this book, we developed a software simulation of a light switch. In Chapter 1, we first implemented a light switch using procedural code, and then we rewrote it using a class. In both cases, the position (or